FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

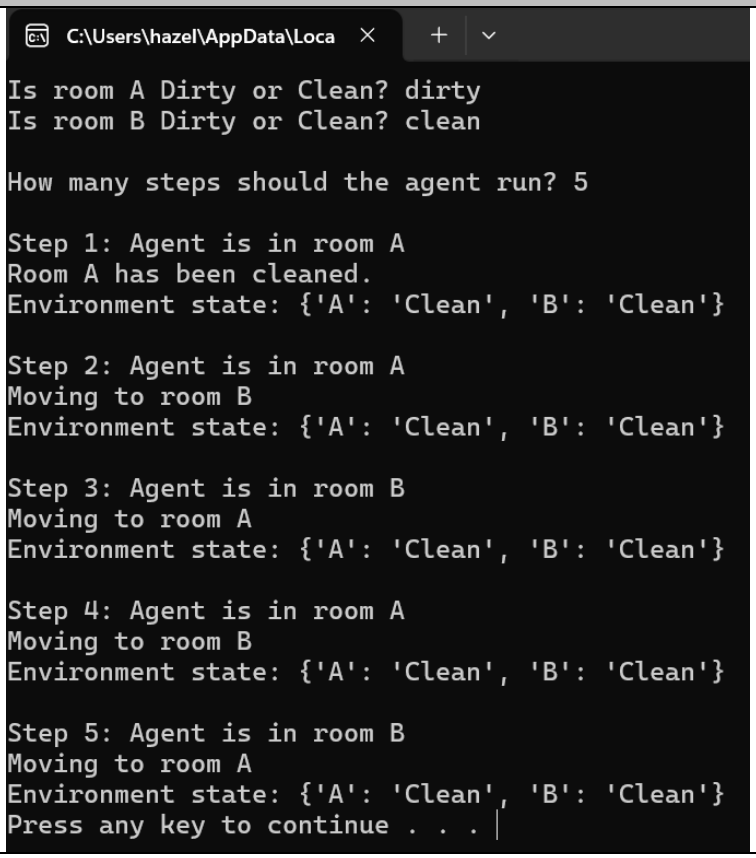
Pre-Lab Preparation

1. Review the Python concepts:
    - Classes and objects
    - If-Else statements
    - Loops (for loop)
    - Functions / methods
2. Understand the environment:
    - Two rooms: A and B
    - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
    - Rule 1: If the current room is dirty, clean it.

- Rule 2: If the current room is clean, move to the other room.

3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.
3. After each step, print:
   - The agent's current location
   - The action taken (cleaned/moved)
   - Current state of the environment

Your Code - Laboratory Tasks ⇩

```
"""Rule-based vacuum agent for the two-room environment (Rooms A and B)."""


def get_room_state(room_name):
    """Ask for a room's starting condition and accept Dirty or Clean."""
    while True:
        answer = input(f"Initial state of Room {room_name} (dirty/clean): ").strip().lower()
        if answer in ("dirty", "clean"):
            return answer
        print("Please enter 'dirty' or 'clean'.")


def is_dirty(environment, room):
    """Return True when the specified room is dirty."""
    return environment[room] == "dirty"


def clean_room(environment, room):
    """Change a room's state to clean."""
    environment[room] = "clean"


class VacuumAgent:
    """A rule-based agent that cleans its current room or moves to the other."""

    def __init__(self, starting_room="A"):
        self.location = starting_room

    def move(self):
        """Switch between Room A and Room B."""
        self.location = "B" if self.location == "A" else "A"
```

```
    def perceive_and_act(self, environment):
        """Apply the rules and return a description of the action taken."""
        if is_dirty(environment, self.location):
            clean_room(environment, self.location)
            return f"Cleaned Room {self.location}"

        old_location = self.location
        self.move()
        return f"Moved from Room {old_location} to Room {self.location}"


def main():
    print("=== Rule-Based Vacuum Agent ===")
    print("Set the initial condition of each room, then run the simulation.\n")

    environment = {
        "A": get_room_state("A"),
        "B": get_room_state("B"),
    }

    while True:
        try:
            steps = int(input("How many simulation steps should run? ").strip())
            if steps < 0:
                print("Please enter 0 or a positive whole number.")
                continue
            break
        except ValueError:
            print("Please enter a whole number, such as 5.")

    agent = VacuumAgent(starting_room="A")
    print("\nStarting environment:", environment)
    print(f"Agent starts in Room {agent.location}.\n")

    for step in range(1, steps + 1):
        action = agent.perceive_and_act(environment)
        print(f"Step {step}:")
        print(f"  Agent location: Room {agent.location}")
        print(f"  Action: {action}")
        print(f"  Environment: {environment}\n")

    print("Simulation complete.")
    print("Final environment:", environment)
```

```
    print(f"Final agent location: Room {agent.location}")


if __name__ == "__main__":
    main()
```

```
PS C:\Users\Robie Dela Cruz>
PS C:\Users\Robie Dela Cruz> py "C:\Users\Robie Dela Cruz\Downloads\TA2_Rule_Based_Agent.py"
=== Rule-Based Vacuum Agent ===
Set the initial condition of each room, then run the simulation.

Initial state of Room A (dirty/clean): dirty
Initial state of Room B (dirty/clean): dirty
How many simulation steps should run? 5

Starting environment: {'A': 'dirty', 'B': 'dirty'}
Agent starts in Room A.

Step 1:
  Agent location: Room A
  Action: Cleaned Room A
  Environment: {'A': 'clean', 'B': 'dirty'}

Step 2:
  Agent location: Room B
  Action: Moved from Room A to Room B
  Environment: {'A': 'clean', 'B': 'dirty'}

Step 3:
  Agent location: Room B
  Action: Cleaned Room B
  Environment: {'A': 'clean', 'B': 'clean'}

Step 4:
  Agent location: Room A
  Action: Moved from Room B to Room A
  Environment: {'A': 'clean', 'B': 'clean'}

Step 5:
  Agent location: Room B
  Action: Moved from Room A to Room B
  Environment: {'A': 'clean', 'B': 'clean'}

Simulation complete.
Final environment: {'A': 'clean', 'B': 'clean'}
Final agent location: Room B
PS C:\Users\Robie Dela Cruz>
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?

   The agent follows two rules. If the room it is currently in is dirty, it cleans that room. If the current room is clean, it moves to the other room.

2. How does the agent decide when to clean and when to move?

   At each step, the agent checks the state of its current room. It cleans when the state is dirty. When the state is clean, it switches its location between Room A and Room B.

3. What happens if both rooms are already clean at the start?

   The agent does not clean either room. Since its current room is clean, it moves to the other room at every step and continues until the number of steps entered by the user is complete.

4. How would you modify the agent to handle **more than two rooms**?

   I would store room names and states in a dictionary or list, such as A, B, and C. The agent could move to the next room in the list after finding its current room clean, returning to the first room after the last one. It would check and clean each room using the same rules.

Answer the Questions ⇩